

La intersección de ideales es un ideal

Ejercicio 1. Sea $\{I_\lambda\}_{\lambda \in \Lambda}$ una colección de ideales de un anillo R . Demuestra que

$$\bigcap_{\lambda \in \Lambda} I_\lambda \text{ es un ideal de } R.$$

Solución: Comprobemos las propiedades de la `ideal`/Definición 1:

- (i) $0 \in \bigcap_{\lambda} I_\lambda$ porque $0 \in I_\lambda$ para todo $\lambda \in \Lambda$.
- (ii) Sean $a, b \in \bigcap_{\lambda} I_\lambda$, entonces $\forall \lambda \in \Lambda : a, b \in I_\lambda$. Como I_λ es un ideal, $\forall \lambda \in \Lambda : a + b \in I_\lambda$ y concluimos que $a + b \in \bigcap_{\lambda} I_\lambda$.
- (iii) Sea $a \in \bigcap_{\lambda} I_\lambda$ y $r \in R$, entonces $\forall \lambda \in \Lambda : a \in I_\lambda$ y por la propiedad de absorción, $\forall \lambda \in \Lambda : ra \in I_\lambda$. Por tanto, $ra \in \bigcap_{\lambda} I_\lambda$.

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